

RICHARD TANG

Manufacturing Engineer in Training

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EDUCATION

University of British Columbia | B.A.Sc. - Mechanical - Manufacturing Engineering

TECHNICAL WORK EXPERIENCE

MDA Space | Brampton, ON | *Manufacturing Engineering Co-op* 05/2024 - 08/2025

- Led continuous improvement initiatives to eliminate non-value-added tasks and enhance workflow efficiency, generating **~\$450K in annualized savings**.
- Developed an internal Procurement Status Tool to streamline inventory management and increase visibility within the MRP system, reducing **procurement cycle time by ~20 hours**.
- Implemented **5S+1 practices** across all engineering labs and machine shops to standardize **80,000+ sq. ft.** of operational space, ensuring consistent safety and productivity standards.
- Authored **200+** detailed Work Instructions (WIs) and Bills of Materials (BOMs) to standardize manufacturing processes, minimizing production variability and ensuring assembly consistency.

SAPA Technologies Ltd. | Vancouver, BC | *Technician* 06/2021 - 08/2021

- Oversaw the assembly of flexible LED light sheets, exceeding production targets of **1,000+ units weekly**.
- Performed quality assurance testing for **THT soldering on 100+ PCBs** for production assemblies, maintaining a **99.8% first-pass yield rate** and ensuring reliable electrical connections.
- Aided in supply chain management by conducting three-way checks and coordinating logistics with shipping providers, maintaining **100% stock accuracy** and managing component inventory flows.

TECHNICAL PROJECTS

3D Printing Semiconductors (Capstone Project) 09/2025 - 04/2026

- Engineered a reliable microscale TEOS deposition system for **semiconductor additive manufacturing** by designing a custom soldered perf board, a protective 3D printed enclosure, and an upgraded **graphical user interface** for precise multi-axis control.
- Validated the synthesis of **silicon dioxide** lines through systematic contact angle, flow rate, and thermal annealing testing using a 10 micron glass micropipette nozzle.

UBC Rocket | *Composite Pressure Vessel Member* 09/2022 - 09/2025

- Fabricated **suborbital rocket endcaps** using woven carbon fiber and epoxy resin with high strength-to-weight ratios, ensuring structural integrity under flight loads.
- Created comprehensive **standard operating procedures** to standardize the manufacturing and layup of composite components

Additive Manufacturing Design 09/2025 - 12/2025

- Redesigned a mechanical assembly using **topology optimization** and optimized print orientation and support structures, minimizing material usage, stress flow, and post-processing time while ensuring manufacturability.

RC Car Design for Manufacturing 09/2023 - 05/2024

- Conducted **mold flow simulation** and heat dissipation analysis on aluminum wheel and chassis designs to validate component geometry and minimize defects.
- Designed a remote-controlled car assembly utilizing injection molding, thermoforming, and CNC machining processes, enabling a hypothetical mass production run of **20,000 annual units**.

S.A.M.I Vertical Farm (Design Competition) | *Project Lead* 03/2023 - 03/2023

- Integrated hardware sensors with a **custom mobile app UI**, enabling real-time plant health monitoring.
- Led a multidisciplinary team using **Arduino** and laser-cut acrylic components to build a "Semi-Autonomous Modular Indoor" farm prototype.

TECHNICAL SKILLS

Design and Modelling: SolidWorks, Siemens NX, Fusion 360, Blender, Figma

Simulation and Analysis: ANSYS (FEA), ABAQUS, MACHPRO, CUTPRO, MATLAB

Manufacturing Processes: CNC Machining, Waterjet, SMT Soldering, MRP Systems, 5S Methodology

Programming and Software: Python, C#, Unity